

Paper 1:AI Driven Fraud Detection in Financial Ecosystems: A Hybrid Machine Learning Framework (2025)

Item	Details
Year	2025
Harvard Citation	Sariat, A.F., Siddique, I.J., Hossain, M., Islam, M.M. and Rahman, T. (2025). <i>AI Driven Fraud Detection in Financial Ecosystems: A Hybrid Machine Learning Framework</i> . IEEE ECCE 2025.
Keywords	Mobile Money, Fraud Detection, Machine Learning, XGBoost, Random Forest, Ensemble Learning
Dataset Used	PaySim Synthetic Mobile Money Dataset (Kaggle)
Methods / Models Used	XGBoost, Random Forest, Logistic Regression, Hybrid Ensemble (Soft Voting)
Best Model	Random Forest (AUPRC = 0.9998)
Research Trend	Ensemble Learning, Feature Engineering, SMOTE, Explainable AI, Real-time Fraud Detection
Limitations	Uses a synthetic dataset; requires validation using real-world financial transaction data.
Research Gap	Future research should focus on Explainable AI, Deep Learning (RNN/LSTM), Privacy-Preserving Machine Learning, Adaptive Learning, and Real-world dataset validation.

Paper 2: An Advanced Hybrid Strategy for Detecting Fraud in Mobile Money Services (2025)

Item	Details
Year	2025
Harvard Citation	Thar, K.W. and Wai, T.T. (2025) <i>An Advanced Hybrid Strategy for Detecting Fraud in Mobile Money Services</i> . IEEE TENCON 2025.
Keywords	Fraud Detection, HMM, XGBoost, SMOTE, Mobile Money, Hybrid ML.
Dataset Used	PaySim; 6.3M+ transactions, 8,213 fraud cases (0.13%).
Methods / Models Used	HMM, XGBoost, SMOTE, Grid Search, SHAP.
Best Model	Hybrid HMM + XGBoost (ROC-AUC: 0.9988 , PR-AUC: 0.9689).
Research Trend	Hybrid ML with sequential feature extraction for fraud detection.
Limitations	Uses only a synthetic dataset; no real-world validation.
Research Gap	Test on real mobile money data and compare with advanced deep learning models.

Paper 3: An Introduction to Machine Learning Methods for Fraud Detection (2025)

Item	Details
Year	2025
Harvard Citation	Compagnino, A.A., Maruccia, Y., Cavuoti, S., Riccio, G., Tutone, A., Crupi, R. and Pagliaro, A. (2025). <i>An Introduction to Machine Learning Methods for Fraud Detection</i> . <i>Applied Sciences</i> , 15(21), 11787.
Keywords	Machine Learning, Deep Learning, Fraud Detection, Data-driven Finance.
Dataset Used	Multiple datasets including Credit Card Fraud , PaySim , BankSim , German Credit , and real banking datasets .
Methods / Models Used	Random Forest, XGBoost, Logistic Regression, SVM, Decision Tree, KNN, ANN, LSTM, CNN, GAN, Isolation Forest, Autoencoder, Ensemble Methods.
Best Model	Random Forest showed the best overall performance in the case study (AUPRC = 0.619).
Research Trend	Hybrid, ensemble and deep learning methods for adaptive fraud detection.
Limitations	Class imbalance, concept drift, privacy issues and limited interpretability.
Research Gap	Develop more adaptive, explainable and privacy-preserving fraud detection models using real-world financial data.

Paper 4: Predicting Mobile Money Transaction Fraud Using Machine Learning Algorithms (2023)

Item	Details
Year	2023
Harvard Citation	Lokanan, M.E. (2023). <i>Predicting Mobile Money Transaction Fraud Using Machine Learning Algorithms</i> . <i>Applied AI Letters</i> , 4(2), e85.
Keywords	Machine Learning, Mobile Money, Money Laundering, Payment Systems.
Dataset Used	PaySim simulated dataset with 1,048,575 mobile money transactions and 9 features , based on real transaction logs from an African mobile money service.
Methods / Models Used	Logistic Regression, Decision Tree, Random Forest, Stochastic Gradient Descent (SGD), SMOTE-ENN.
Best Model	Random Forest (Accuracy: 88% , Precision: 0.96 , Recall: 0.80 , F1-score: 0.87).
Research Trend	Applying machine learning models to detect fraud and money laundering in mobile money transactions.
Limitations	Used a simulated dataset and evaluated only a limited number of ML models; Random Forest may overfit.
Research Gap	Validate using real mobile money data and investigate advanced deep learning and hybrid models for improved fraud detection.

Paper 5: Detection of Anomalous Transactions in Mobile Payment Systems (2020)

Item	Details
Year	2020
Harvard Citation	Hussain, I. and Asif, M. (2020). <i>Detection of Anomalous Transactions in Mobile Payment Systems. International Journal of Data Analytics</i> , 1(2), pp. 58–66.
Keywords	Machine Learning, Fraud Detection, PAYSIM, Financial Data Mining, Classification, Mobile Payment.
Dataset Used	PAYSIM synthetic financial transaction dataset generated using the PaySim simulator due to the lack of real-world data.
Methods / Models Used	Decision Table, JRip (RIPPER), PART, J48 (C4.5), Decision Stump, Naïve Bayes, IBK (KNN), Locally Weighted Learning (LWL).
Best Model	J48 (C4.5) achieved the best performance with 99.55% accuracy, 99.70% precision, 96.81% recall, and 98.23% F1-score .
Research Trend	Comparing multiple machine learning classifiers to improve anomaly detection in mobile payment systems.
Limitations	Used a synthetic dataset and evaluated only traditional machine learning algorithms.
Research Gap	Future work should validate models on real transaction data and investigate deep learning techniques for evolving fraud patterns.

Paper 6: Detection of Financial Fraud: Comparisons of Some Tree-Based Machine Learning Approaches (2023)

Item	Details
Year	2023
Harvard Citation	Sengupta, K. and Das, P.K. (2023). <i>Detection of Financial Fraud: Comparisons of Some Tree-Based Machine Learning Approaches. Journal of Data, Information and Management</i> , 5, pp. 23–37.
Keywords	Financial Fraud, PaySim, Tree-based Learning, SHAP, XGBoost, CatBoost.
Dataset Used	PaySim benchmark dataset with 6.36 million mobile payment transactions and 8,213 fraud cases .
Methods / Models Used	Decision Tree, Random Forest, GBM, AdaBoost, XGBoost, LightGBM, CatBoost, SHAP.
Best Model	XGBoost and CatBoost achieved the best overall performance, with XGBoost obtaining the highest PR-AUC (0.9193) and F1-score (0.9166) .
Research Trend	Comparing advanced tree-based ensemble models with explainable AI (SHAP) for fraud detection.
Limitations	Used the synthetic PaySim dataset and focused only on tree-based machine learning models.

Research Gap	Future research should validate models on real-world transaction data and explore deep learning and hybrid fraud detection approaches.
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Paper 7: Fraud Detection in Mobile Payment Systems using an XGBoost-based Framework (2023)

Item	Details
Year	2023
Harvard Citation	Hajek, P., Abedin, M.Z. and Sivarajah, U. (2023). <i>Fraud Detection in Mobile Payment Systems using an XGBoost-based Framework</i> . <i>Information Systems Frontiers</i> , 25, pp. 1985–2003.
Keywords	Mobile Payment, Fraud Detection, Machine Learning, XGBoost, Imbalanced Data, Outlier Detection.
Dataset Used	PaySim dataset with 6.36 million transactions and 8,213 fraud cases .
Methods / Models Used	RUS + XGBoost, XGBOD, k-NN, SVM, Random Forest, Isolation Forest, LOF, Autoencoder.
Best Model	XGBOD achieved the best overall performance, while RUS + XGBoost provided the highest financial cost savings.
Research Trend	Hybrid and semi-supervised ML for handling imbalanced mobile payment fraud data.
Limitations	Uses the synthetic PaySim dataset and has high computational cost.
Research Gap	Validate on real-world mobile payment data and develop adaptive deep learning models.

Paper 8: Multi-Class Mobile Money Service Financial Fraud Detection by Integrating Supervised Learning with Adversarial Autoencoders (2021)

Item	Details
Year	2021
Harvard Citation	Silva, J.C.S., Macedo, D., Zanchettin, C., Oliveira, A.L.I. and Almeida Filho, A.T. (2021). <i>Multi-Class Mobile Money Service Financial Fraud Detection by Integrating Supervised Learning with Adversarial Autoencoders</i> . In: <i>2021 International Joint Conference on Neural Networks (IJCNN)</i> . IEEE.
Keywords	Fraud Detection, Mobile Money Service, Adversarial Autoencoders, Supervised Learning, Multi-class Classification.
Dataset Used	Synthetic mobile money dataset with 533,009 transactions and 8 attributes (618 features after encoding); classified into regular, global anomaly, and local anomaly .
Methods / Models Used	Adversarial Autoencoder (AAE), Random Forest, SVM, Naïve Bayes, 10-fold Cross-Validation.
Best Model	Random Forest achieved the best performance (MAUC = 0.9972).
Research Trend	Combining deep learning feature extraction with supervised learning for multi-class fraud detection.

Limitations	Uses a synthetic dataset and a single autoencoder architecture; lacks real-world validation.
Research Gap	Evaluate on real mobile money data and explore advanced autoencoder architectures and modern classifiers.

Paper 9: No Smurfs: Revealing Fraud Chains in Mobile Money Transfers (2014)

Item	Details
Year	2014
Harvard Citation	Zhdanova, M., Repp, J., Rieke, R., Gaber, C. and Hemery, B. (2014). <i>No Smurfs: Revealing Fraud Chains in Mobile Money Transfers</i> . In: <i>2014 9th International Conference on Availability, Reliability and Security (ARES)</i> . IEEE.
Keywords	Mobile Money Transfer, Money Laundering, Smurfing, Fraud Chain Detection, Predictive Security Analysis.
Dataset Used	Synthetic dataset with 466,359 transactions , 10,000 users , and 10 fraud chains .
Methods / Models Used	Fraud Chain Detection (FCD), Predictive Security Analysis at Runtime (PSA@R), Random Forest, C4.5, PART Decision Table.
Best Model	FCD (PSA@R) achieved 99.81% Precision and 90.18% Recall , outperforming ML models.
Research Trend	Model-based fraud chain detection for identifying money laundering in mobile money transfers.
Limitations	Uses a synthetic dataset; needs validation on real-world mobile money data.
Research Gap	Test the approach on real transaction datasets and extend it to detect other fraud types in mobile money systems.

Paper 10: Telecommunication Fraud Resilient Framework for Efficient and Accurate Detection of SMS Phishing Using Artificial Intelligence Techniques (2024)

Item	Details
Year	2024
Harvard Citation	Hapase, D.S. and Patil, L.V. (2024). <i>Telecommunication Fraud Resilient Framework for Efficient and Accurate Detection of SMS Phishing Using Artificial Intelligence Techniques</i> . <i>Multimedia Tools and Applications</i> , 83, pp. 89111–89133.
Keywords	Deep Learning, SMS Phishing Detection, PCC-PCA, Enhanced CNN, Fraud Classification.
Dataset Used	UCI SMS Spam Collection v1 with 5,574 SMS messages (4,827 ham, 747 spam) ; 70:30 train-test split.
Methods / Models Used	TF-IDF, Tokenization, Stemming, PCC-PCA, Enhanced CNN, SVM, Naïve Bayes, MNB, KNN.

Best Model	Enhanced CNN achieved 99.8% Accuracy , 99.75% Recall , and 99.0% F1-score .
Research Trend	Deep learning with feature engineering for SMS phishing detection.
Limitations	Evaluated only on the SMS Spam dataset; lacks validation on real telecom fraud data.
Research Gap	Validate on real-world telecom/mobile money datasets and develop adaptive fraud detection models.

Paper 11: Explainable AI-Enhanced Ensemble Learning for Financial Fraud Detection in Mobile Money Transactions (2025)

Item	Details
Year	2025
Harvard Citation	Vijayanand, D. and Smrithy, G.S. (2025). <i>Explainable AI-Enhanced Ensemble Learning for Financial Fraud Detection in Mobile Money Transactions</i> . <i>Intelligent Decision Technologies</i> , 19(1), pp. 52–67.
Keywords	Explainable AI, SHAP, Ensemble Learning, Voting Classifier, Mobile Money, Fraud Detection.
Dataset Used	PaySim dataset with 6.36 million transactions and 8,213 fraud cases (0.13%) .
Methods / Models Used	Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, Neural Network, Voting Classifier, SHAP.
Best Model	Voting Classifier (Ensemble Learning) with 99.904% Accuracy and 0.990 ROC-AUC .
Research Trend	Combining Explainable AI (SHAP) with ensemble learning for transparent fraud detection.
Limitations	Uses a synthetic PaySim dataset and lacks real-world validation.
Research Gap	Validate on real mobile money data and develop real-time adaptive fraud detection models.